Project #1: Materials Tracking and Order Handling

Learning Outcomes:

The goal of this project is to provide an opportunity to use what we've learned about Excel and spreadsheets in class to design a simple tool for handling orders, managing materials, and communicating information between different parts of a production operation. A small group of 2 to 3 (depending on size of the class) will work to develop this tool while practicing effective teamwork and technical communication while presenting their project in a demonstration to stakeholders.

Background:

Your engineering services company has been asked to bid on a job to develop a custom materials tracking and invoicing system for a small furniture manufacturing company. As part of the sales pitch to the client you will develop and demonstrate a proof-of-concept spreadsheet that includes the major features they are looking for in the system.

Project Goals:

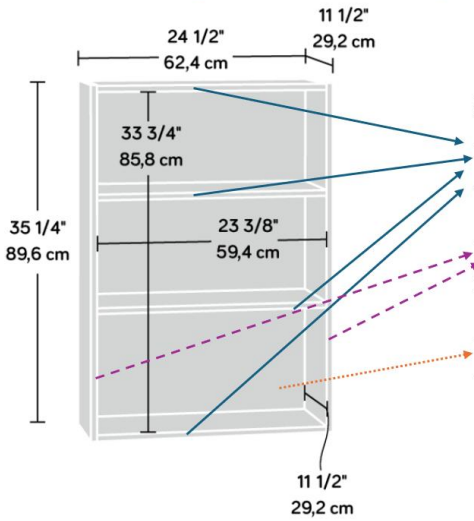
Your team will create a prototype spreadsheet workbook that will use a database of catalog information and order input from the user to create both an invoice of the order for the shipping and accounting department, and a materials list for the warehouse and production facility. Your complete project should consist of an Excel workbook with three spreadsheets: **Catalog Data**, **Invoicing**, and **Materials**.

The **Catalog Data** sheet should contain all the information about each of the different furniture pieces that could be ordered. This spreadsheet will serve data to the other spreadsheets that will make up this project. Take the sample catalog provided by the company (catalog.pdf) and create a table that contains information about the price, description, weight, and components required to manufacture each item.

Part Number: 428232

- Assume all sides, top, bottom, and shelves are made of ¾" material.
- Assume all backing plywood is 0.1875" thick.
- Do not worry about trim, facing, or legs/supports for this demonstration.





Component Breakdown

The shelves and top:
4 pieces 24.5" x 11.5" x 0.75"

The sides:
2 pieces 35.25" x 11.5" x 0.75"

The backing plywood:
1 piece 35.25" x 24.5" x 0.1875"

Price: $49.99
Small 3-Shelf Bookcase: Silver
Weight: 33.9 lbs.

The **Catalog Data** sheet will have cut lists (a list of quantities and sizes for each of the wooden panels used to make the furniture) for each item. You will have to decide how to arrange these, but it may make sense to organize them by thickness as shown in the example below. Within each thickness group the parts are arranged by their largest dimension first. Note that any item with a component that has a single dimension greater than 48" or two dimensions greater than 30" will incur extra shipping costs as "oversized."

*Indicates oversized item for shipping, any one dimension over 48" or any two dimensions over 30")

					0.75" thick			0.1875" thick	
Item No.	Price	Description	Weight [lbs]	*Oversize	*69.75" x 13.25	35.25" x 11.5"	24.5" x 11.5"	*69.75" x 35.25"	35.25" x 24.5"
428232	$49.99	Small 3-Shelf Bookcase: Silver	33.9	no		2	4		1
401635	$72.49	Small Underdesk shelf: Black	28.5	no		1	2		
382512	$115.99	Large display case: Red Oak	64.2	yes	2	3		1	

The **invoicing** sheet should display a table containing <u>at minimum</u> the: *Item number*, *quantity*, *item description*, *unit cost*, and *total cost*. Below the table should be the *total cost of goods* (sum of all items being purchased) as well as *shipping costs* (basic shipping and *oversize surcharge* if applicable) and an *order total*. For example:

Order Number	**812**
Order Date	10/15/2024

Item Number	Quantity	Part Number	Description	Unit Cost	Total
1	2	428232	Small 3-Shelf Bookcase: Silver	$49.99	$99.98
2					
...					
...					
...					
...					
...					
...					
...					
10					

Item Total		**$99.98**
Shipping		$27.76
Oversize	NA	
Order Total		**$127.74**

The **invoicing** sheet should automatically use information from the Catalog Data sheet, so that when the user enters the quantity and part number the description and unit cost data are pulled from the catalog.

Shipping costs should be calculated as follows:

> There is a flat $12.00 fee for all shipments. Additionally, there is a $0.25 per pound charge for the first 40 pounds and $0.20 per pound for anything over that first 40 pounds. If anything in the order is oversized the flat fee increases to $35.

For example, in the invoice shown above the total weight of the order is 67.8 pounds, and the order is not oversized. In this case the shipping costs would be: $12.00 + 40lbs (0.25/lb) + 27.8lbs (0.20/lb) = $27.76.

The **Materials** spreadsheet will take *quantity* and *part number* data from the **Invoicing** spreadsheet and then find the necessary information in the **Catalog Data** spreadsheet to create a list of components required to complete the order. The **Materials** sheet should provide those in the production / warehouse facilities with the number and size of all the panels needed to fulfil the order. In addition, the amount of materials consumed (measured in board-feet, where 1 board-foot = 144 cubic inches) should be calculated for each panel size. These individual material amounts should then be totaled up to determine the overall material requirement. A sample of a possible **Materials** sheet format is shown below:

Order Number	812
Order Date	10/15/2024

Total Materials Required	
49.20	[board-feet]

		0.75" Thick Material			0.1875" Thick Material	
Qty.	Item No.	*69.75" x 13.25	35.25" x 11.5"	24.5" x 11.5"	*69.75" x 35.25"	35.25" x 24.5"
2	428232	0	4	8	0	2
	Total	0	4	8	0	2
Material volume [board feet]		0.000	8.445	38.508	0.000	2.249
(1 board-foot = 144 cubic inches)						

Your spreadsheets should make use of the functions and techniques we have discussed in class when implementing the requested features.

- It might be tempting to put everything on a single spreadsheet but remember that you're creating the demo from a small subset of your client's data. You want your proof-of-concept to reflect what the final product would look like.
- The order information on the **Invoicing** sheet and the majority of the **Catalog Data** spreadsheet are the only parts of the workbook that should be 'hard coded' meaning values are typed in and fixed rather than calculated or derived from elsewhere.
- Plan on making use of VLOOKUP along with MATCH and INDEX functions to pull the required data between sheets.
- Data Validation should be used to limit user inputs to reasonable values (for example, users should not be able to order and item number that doesn't appear in the catalog).
- Use of the IF, AND, OR, and IFERROR functions along with Conditional Formatting can help create a good user experience.
- Adding color coding or other visual elements can improve your spreadsheet but avoid eye-straining color schemes and overuse of color. Using cell shading and font effects should help focus the user, but too much can be confusing.

Deliverables:

The deliveries for this project include:

1. A sample Excel workbook that has been developed as a group and contains the features outlined above. Your spreadsheet will be submitted via Canvas prior to your team's demonstration.

2. A demonstration of your solution where your team will demonstrate the operation of the spreadsheet and be prepared to answer questions about its design, features, and operation. Your team will have 10 minutes to demonstrate your spreadsheet's operation and answer questions about its development.

   Each team member will demonstrate the portion of the project that they worked on and should be prepared to discuss how they accomplished the features of that portion of the project (Specific Excel function use, formatting, and data validation).

Grading:

The grading for this project will be based on the two deliverables with 30 points allotted to the Excel workbook containing your proof-of-concept, and 35 points based on the demonstration of your proposed solution to the client.

The workbook document will be graded as a team effort with each group member receiving the same score.

The presentation portion of the assignment will be graded individually, with the grade based on the team members communication of their portion of the project and knowledge of its development.

Additionally, there will be a short post-project feedback form that will be worth 5 points.

Rubric:

For the Excel workbook I will be looking at the following criteria:

- The workbook should be well organized with separate sheets for the **Invoice**, **Materials**, and **Catalog Data** sections.
- The **Invoice** sheet should allow the user to add part numbers and quantities, and then pull all related data dynamically from the **Catalog Data** sheet. Data validation should be used so only positive quantities may be entered and only valid part numbers can be chosen.
- The **invoice sheet** should also calculate the shipping for the order based on total order weight and whether or not oversized components are part of the order.
- The **Catalog Data** sheet should contain information about the number and dimensions of the components required to make up each furniture piece. There are many ways to arrange this, the important part is that it is well organized and can serve data to the other spreadsheets in the workbook.
- The **Materials** spreadsheet should pull quantity and item numbers from the **Invoice** sheet, and then get the relevant information (how many of each size of panel) from the **Catalog Data** sheet.
- The total number of board feet for each size of panel should be calculated, as well as the total board feet for the entire order.
- Use of color coding or other graphical techniques should be used to highlight which panel sizes are required for the order.

For the presentation I will be looking at the following:

- The group should give a quick overview and basic demonstration of the operation of their spreadsheet: For example, showing how the client could add and remove items from the **invoice** sheet and how that will generate a corresponding **materials** sheet.
- The group should emphasize features and design elements that make their product intuitive and easy to use.
- Each member should be prepared to talk about the technical details (use of functions, structure of the spreadsheet) of the section of the project they were responsible for.
- Since we are simulating pitching a demon to a client, I'll be looking for your presentation to be smooth with good coordination between team members. It doesn't have to be perfect, but you should be able to talk about your product easily.